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Computational Experience with Known Variable Metric Updates

L. LUKŠAN[1]

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Abstract. Variable metric methods from the Broyden family are well known and commonly used for unconstrained minimization. These methods have good theoretical and practical convergence properties which depend on a selection of free parameters. We demonstrate, using extensive computational experiments, the influence of both the Biggs stabilization parameter and Oren scaling parameter on 12 individual variable metric updates, two of which are new. This paper focuses on a class of variable metric updates belonging to the so-called preconvex part of the Broyden family. These methods outperform the more familiar BFGS method. We also experimentally demonstrate the efficiency of the controlled scaling strategy for problems of sufficient size and sparsity.

Key Words. Quasi-Newton updates, Broyden family, variable metric methods, unconstrained optimization, nonlinear programming.

1. Introduction

Variable metric methods are the most frequently used optimization methods for unconstrained problems of a moderate size. These methods are iterative and their iteration step has the form

$$x^{+} = x + \alpha s, \tag{1}$$

where

$$s = -Hg. \tag{2}$$

[1]Researcher, Institute of Computer Science, Academy of Sciences of the Czech Republic, Prague, Czech Republic.

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Here, x and x^+ are old and new vectors of variables respectively, α is a positive stepsize chosen so that

$$F^+ - F \leq \epsilon_1 \alpha s^T g, \tag{3}$$

$$s^T g^+ \geq \epsilon_2 s^T g, \tag{4}$$

with $0 < \epsilon_1 < 1/2$ and $\epsilon_1 < \epsilon_2 < 1$, where F and F^+ are the old and new values of the objective function respectively, g and g^+ are the old and new gradients of the objective function respectively, and H is a symmetric positive-definite approximation of the inverse Hessian matrix that is constructed by means of the formula

$$\begin{aligned} H^+ = \gamma\{H + (\rho/\gamma)(1/b)dd^T - (1/a)Hy(Hy)^T \\ + (\eta/a)[(a/b)d - Hy][(a/b)d - Hy]^T\}, \end{aligned} \tag{5}$$

with three parameters $\rho > 0$, $\gamma > 0$, and η, which are chosen at least in the way that H^+ is also a symmetric positive-definite matrix. In (5) and in the subsequent text, we use the notation

$$d = x^+ - x = \alpha s, \tag{6}$$

$$y = g^+ - g, \tag{7}$$

$$a = y^T H y, \qquad b = y^T d, \qquad c = d^T H^{-1} d, \tag{8}$$

$$\hat{a} = a/b, \qquad \hat{c} = c/b, \tag{9}$$

and we suppose that $b = y^T d > 0$, as follows from (4). Then, the matrix H^+ is positive definite if and only if $\eta > \eta^*$, where

$$\eta^* = -1/(\hat{a}\hat{c} - 1) < 0 \tag{10}$$

is a degenerate value of the parameter η. Note that, for positive definite H, we have $a > 0$, $c > 0$, and, therefore, $\hat{a} > 0$, $\hat{c} > 0$. Note also that, from the Schwartz inequality, $\hat{a}\hat{c} - 1 \geq 0$ holds with strict inequality for linearly independent vectors d and Hy.

If we set

$$B = H^{-1} \quad \text{and} \quad B^+ = (H^+)^{-1},$$

we can write (5) in the form

$$\begin{aligned} B^+ = (1/\gamma)\{B + (\gamma/\rho)(1/b)yy^T - (1/c)Bd(Bd)^T \\ + (\beta/c)[(c/b)y - Bd][(c/b)y - Bd]^T\}, \end{aligned} \tag{11}$$

where

$$\eta\beta(\hat{a}\hat{c} - 1) + \eta + \beta = 1. \tag{12}$$

Again, the matrix B^+ is positive definite if and only if $\beta > \beta^*$, where

$$\beta^* = -1/(\hat{a}\hat{c} - 1) < 0 \tag{13}$$

is a degenerate value of the parameter β.

More details about variable metric methods can be found in Refs. 1 and 2.

The main purpose of this work is a presentation of numerical properties of known variable metric methods. This work contains no theoretical results concerning convergence analysis, but the computational results presented here are very interesting and they can challenge widely held notions on variable metric methods. After describing various approaches for choice of parameters $\rho > 0$, $\gamma > 0$, and η, we describe in detail the implementation of variable metric methods used for testing and show their properties in several tables. The tests were done using 25 sufficiently difficult problems taken from Refs. 3 and 4 with different number of variables.

2. Selection of Stabilization and Scaling Parameters

The stabilization parameter $\rho > 0$ was introduced by Biggs in Ref. 5. It was used for stabilization of variable metric methods in the case of strongly nonquadratic functions. Let us consider the update (5). Then, we can write

$$H^+ y = \rho d, \tag{14}$$

which is a relaxed quasi-Newton condition (standard quasi-Newton condition uses $\rho = 1$, which corresponds to a quadratic objective function). Define

$$\phi(\lambda) = F(x + \lambda d).$$

The parameter ρ should be chosen in such a way that

$$d^T (H^+)^{-1} d = \phi''(1)$$

holds, where $\phi''(\lambda)$ is the second derivative of the function $\phi(\lambda)$. Using (14), we can see that this condition is satisfied for the value

$$\rho = [\phi'(1) - \phi'(0)]/\phi''(1), \tag{15}$$

where $\phi'(1) - \phi'(0) = b$; see (8). Let

$$P = (\phi(1) - \phi(0))/\phi'(0) \quad \text{and} \quad Q = \phi'(1)/\phi'(0).$$

The ratio (15) can be determined using some approximation of the function $\phi(\lambda)$:

(A1) Homogeneous Approximation. Biggs used the homogeneous approximation

$$\phi(\lambda) \approx c_2|\lambda - c_1|^r + c_0$$

in Ref. 5. This choice does not have practical importance, since a difficult nonlinear equation has to be solved. We propose a simpler model $\phi(\lambda) \approx c_2\lambda^r + c_1\lambda + c_0$, for which the exponent r can be easily eliminated, yielding

$$\rho = (P - 1)/(Q - P). \tag{16}$$

(A2) Cubic Approximation. Biggs used a cubic approximation of the form

$$\phi(\lambda) \approx c_3\lambda^3 + c_2\lambda^2 + c_1\lambda + c_0$$

in Ref. 6. In this case, (15) implies

$$\rho = (1/2)(Q - 1)/(2(Q - 1) - 3(P - 1)). \tag{17}$$

(A3) Backward Taylor Expansion. Using the backward Taylor expansion, we obtain

$$\phi(0) = \phi(1) - \phi'(1) + (1/2)\phi''(\lambda),$$

where $0 \leq \lambda \leq 1$. If we approximate

$$\phi''(1) \approx \phi''(\lambda)$$

and use (15), then we obtain

$$\rho = (1/2)(Q - 1)/(Q - P). \tag{18}$$

This formula was discovered by Spedicato in Ref. 7 and was also studied by Yuan in Ref. 8.

We have also tested some other possibilities, but the choice (18) was shown to be best. This formula can be written in the form

$$\rho = (1/2)d^T y/(F - F^+ + d^T g^+), \tag{19}$$

which is slightly more stable than (18) due to roundoff errors. The choice of ρ has to be safeguarded. In this case, we first compute the ratio (19); if this ratio does not lie in the interval $[10^{-2}, 10^{+2}]$, we set $\rho = 1$.

The scaling parameter $\gamma > 0$ was introduced by Oren and Luenberger in Ref. 9. The motivation for scaling and the determination of the scaling parameter $\gamma > 0$ can be explained in three different ways:

(S1) Heuristic Scaling. Since the matrix H^+ has to satisfy the relaxed quasi-Newton condition (14), we can choose the parameter $\gamma > 0$ in such a way that $\gamma Hy \approx \rho d$. This leads to a smaller change from γH to H^+. If we premultiply $\gamma Hy \approx \rho d$ by the transpose of y, we obtain

$$\gamma/\rho = b/a. \tag{20}$$

Similarly, if we use $H^{-1}d$ instead of y, we get

$$\gamma/\rho = c/b. \tag{21}$$

Another useful value is the geometric mean

$$\gamma/\rho = (c/a)^{1/2}. \tag{22}$$

(S2) Optimally Conditioned Scaling. This approach was introduced by Oren and Spedicato in Ref. 10 and it was also studied by Lukšan in Ref. 3 and by Hu and Storey in Ref. 11. The motivation for optimally conditioned scaling is the minimization of the spectral condition number $\kappa(H^{-1/2}H^+H^{-1/2})$. It can be shown (see Ref. 11) that, if

$$1-(1-\eta^*)^{1/2} \leq \eta \leq 1+(1-\eta^*)^{1/2}, \tag{23}$$

then the minimum of $\kappa(H^{-1/2}H^+H^{-1/2})$ is obtained for

$$\gamma/\rho = \hat{c}/(1+\eta(\hat{a}\hat{c}-1)). \tag{24}$$

If (23) is not satisfied, then the minimum of $\kappa(H^{-1/2}H^+H^{-1/2})$ is obtained for

$$\gamma/\rho = (1-1/\eta)/\hat{a},$$

the rank-one formula. Note that most variable metric updates studied in the next section satisfy the condition (23). For the BFGS method, we have

$$\eta = 1 \quad \text{and} \quad \gamma/\rho = 1/\hat{a}.$$

For the DFP method, we have

$$\eta = 0 \quad \text{and} \quad \gamma/\rho = \hat{c}.$$

(S3) Variationally Derived Scaling. This approach was introduced by Al-Baali and Fletcher in Ref. 12 and it was also studied by Lukšan in Ref. 13. The motivation for variationally derived scaling is a minimization of the Frobenius norm $\|(B^+)^{1/2}(H^+-\gamma H)(B^+)^{1/2}\|_F$. It can be shown (see Ref. 13) that, if

$$\eta > (\hat{a}\hat{c}-2)/(\hat{a}\hat{c}-1), \tag{25}$$

then the minimum of $\|(B^+)^{1/2}(H^+-\gamma H)(B^+)^{1/2}\|_F$ is obtained for

$$\gamma/\rho = [1-(\hat{a}\hat{c}-1)/(1+\eta(\hat{a}\hat{c}-1))]/\hat{a}. \tag{26}$$

For the BFGS method, we have

$$\eta = 1 \quad \text{and} \quad \gamma/\rho = 1/(\hat{a}^2\hat{c}).$$

For the DFP method, we have

$$\eta = 0 \quad \text{and} \quad \gamma/\rho = (2 - \hat{a}\hat{c})/\hat{a},$$

which can be used only if $\hat{a}\hat{c} < 2$.

Even if both the optimally conditioned scaling and the variationally derived scaling have some theoretical support, numerical tests show that the heuristic approach is usually better. Moreover, great attention has to be devoted to the scaling strategy. Besides two extremal strategies, either no scaling or scaling in all iterations, we can use some other strategies. The most widely known and commonly used strategy is preliminary scaling introduced by Shanno and Phua in Ref. 14. This strategy consists in the scaling of matrix H whenever it is reset to be the identity matrix; it appears either in the first iteration or in the case where the direction vector s is almost orthogonal to the gradient g.

Another strategy, controlled scaling, is based on results of current line searches. As will be shown later, the line search algorithm generates a sequence of values $\alpha_1, \ldots, \alpha_p$ such that $\alpha^+ = \alpha_p$ is the first value satisfying both conditions (3) and (4). The main significance of scaling consists in its influence on the requirement of the initial stepsize acceptability ($\alpha^+ = \alpha_1 = 1$ should be satisfied for most iterations). There are two influences that fight against each other. At the beginning of the iterative process, the scaling usually adapts the initial matrices to be closer to the corresponding Hessians which increases the probability of $\alpha_1 = 1$ being accepted. At the end of the iterative process, the scaling can violate convergence of the final matrices to the corresponding Hessians, so that $\alpha_1 = 1$ may not be a suitable choice. Therefore, a reasonable strategy should not use the scaling whenever the stepsize $\alpha_1 = 1$ gives a result successful in some sense. One such strategy is the controlled scaling proposed in Ref. 13, which is defined by the following algorithm.

Controlled Scaling Algorithm. Given F, $F_1 = F(x + \alpha_1 s)$, F^+, g, $g_1 = g(x + \alpha_1 s)$, g^+, s.

Step 1. Compute a scaling parameter $\gamma > 0$ according to one of the rules given above; see (20)–(26). In the first iteration or after restart, go to Step 6.

Step 2. Compute the ratio $\lambda_1 = s^T g_1 / s^T g$. If $|\lambda_1| \leq 0.2$ and $F_1 \leq F$, then set $\gamma = 1$.

Step 3. If $\gamma > 1$ and if $F_1 > F$ or $\lambda_1 < 0$, then set $\gamma = 1$.

Step 4. If $\gamma < 1$ and if $F_1 \leq F$ and $\lambda_1 > 0$, then set $\gamma = 1$.

Step 5. If $\gamma < 0.5$ or $\gamma > 2.5$, then set $\gamma = 1$.

Step 6. Update the matrix H by (5) using the scaling parameter $\gamma > 0$ obtained in the previous steps.

As with preliminary scaling, controlled scaling always changes the matrix H whenever it is reset to be the identity matrix. This feature is essential and removing it decreases the performance of the algorithm. While Steps 2–4 can have a logical justification, Step 5 of the controlled scaling algorithm is quite heuristic. Each attempt to change it led to a deterioration of the algorithm efficiency. Even after devoting a great deal of effort to finding other efficient scaling strategies, the controlled scaling algorithm given above was not outperformed.

3. Variable Metric Updates

Variable metric methods can be realized in several ways. The most efficient implementation, measured by computational time, is the iteration (1)–(5) with standard line search. Let us denote for simplicity

$$\hat{\rho} = \rho/\gamma, \qquad \hat{\gamma} = \gamma/\rho.$$

The formula (5) can be written in the more economical form with two rank-one corrections,

$$H^+ = \gamma\{H + (\omega/b)[d - (\eta/\omega)Hy][d - (\eta/\omega)Hy]^T + (1/a)(\eta\hat{\rho}/\omega - 1)Hy(Hy)^T\}, \tag{27}$$

where $\omega = \eta\hat{a} + \hat{\rho}$. We use this general formula when the special and simpler one cannot be derived. Individual variable metric methods are distinguished by the parameter η in (27). If $\eta \leq \eta^*$, then the matrix (27) is not positive definite and the variable metric method is not well defined. If $\eta^* < \eta < 0$, then a reasonable convergence theory does not exist and computational tests also indicate low efficiency. If $0 \leq \eta \leq 1$, then we obtain the most widely known subclass of variable metric updates, the so-called convex part of the Broyden family. If $\eta \geq 1$, then we obtain the so-called preconvex part of the Broyden family; we also use the simpler name "preconvex" for these variable metric updates. The last two subclasses, defined by the inequality $\eta \geq 0$, contain efficient methods which will be studied here. We concentrate our attention on the following variable metric updates:

(U1) BFGS Update. See Refs. 15–18. This update is defined by the choice $\eta = 1$. In this case, $\omega = \hat{a} + \hat{\rho}$ in (27). The optimally conditioned scaling parameter is $\hat{\gamma} = 1/\hat{a}$. The recommended heuristic scaling parameter is $\hat{\gamma} = \hat{c}$.

(U2) DFP Update. See Refs. 19 and 20. This update is defined by the choice $\eta = 0$. In this case,

$$H^+ = \gamma\{H + \hat{\rho}(1/b)dd^T - (1/a)Hy(Hy)^T\}.$$

The optimally conditioned and recommended heuristic scaling parameter is $\hat{\gamma} = \hat{c}$.

(U3) Hoshino Update. See Ref. 21. This update is defined by the choice

$$\eta = 1/(1 + \hat{\gamma}\hat{a}). \tag{28}$$

In this case,

$$H^+ = \gamma\{H + (2\hat{\rho}/b)dd^T - (1/b)[1/(\hat{\rho} + \hat{a})](\hat{\rho}d + Hy)(\hat{\rho}d + Hy)^T\}.$$

The optimally conditioned and recommended heuristic scaling parameter is $\hat{\gamma} = (\hat{c}/\hat{a})^{1/2}$. If we set this value into (28), we obtain

$$\eta = 1/[1 + (\hat{a}\hat{c})^{1/2}]. \tag{29}$$

This choice, proposed by Oren and Spedicato in Ref. 10 and studied by Yuan in Ref. 22, is different from the choice (28) if $\hat{\gamma}$ is not given by (22). All other choices derived from the expression (28), some of them given in Ref. 23, satisfy also the condition $0 \le \eta \le 1$ and they have similar properties, as has been found by our computational tests. Therefore, we shall consider only the standard Hoshino update.

(U4) Safeguarded Rank-One Update. See Ref. 3. This rank-one update is defined by the choice

$$\eta = 1/(1 - \hat{\gamma}\hat{a}). \tag{30}$$

In this case,

$$H^+ = \gamma\{H + (1/b)[1/(\hat{\rho} - \hat{a})](\hat{\rho}d - Hy)(\hat{\rho}d - Hy)^T\}.$$

Since the condition $\eta \ge 0$ is satisfied only if $\hat{\gamma} < 1/\hat{a}$, the rank-one update has to be safeguarded. A reasonable strategy consists in the choice (30) whenever $\hat{\gamma} < 1/\hat{a}$ holds and the choice $\eta = 1$ (BFGS method) in the opposite case. The optimally conditioned scaling parameter, determined from (24) and satisfying $\hat{\gamma} < 1/\hat{a}$, is

$$\hat{\gamma} = \hat{c}[1 - (1 - 1/\hat{a}\hat{c})^{1/2}]. \tag{31}$$

It works well for the preliminary scaling, but becomes unstable for more frequent scaling. Therefore, we recommend the heuristic scaling parameter

$\hat{\gamma} = (\hat{c}/\hat{a})^{1/2}$ instead of (31). Note that, if we use this heuristic scaling parameter, then the condition $\hat{\gamma} < 1/\hat{a}$ is not satisfied and we have to use the BFGS update.

(U5) Optimally Conditioned Update. See Ref. 24. This update is defined by the choice

$$\eta = (\hat{\rho}\hat{c} - 1)/(\hat{a}\hat{c} - 1), \qquad \text{if } \hat{\gamma}^2\hat{a} - 2\hat{\gamma}\hat{a}\hat{c} + \hat{c} < 0, \tag{32a}$$

$$\eta = 1/(1 - \hat{\gamma}\hat{a}), \qquad \text{if } \hat{\gamma}^2\hat{a} - 2\hat{\gamma}\hat{a}\hat{c} + \hat{c} \geq 0. \tag{32b}$$

This choice is the optimally conditioned update for an arbitrary scaling parameter $\hat{\gamma} > 0$; it follows from (23) and (24). The recommended scaling parameter is $\hat{\gamma} = (\hat{c}/\hat{a})^{1/2}$.

(U6) Preconvex Update Determined from the Rank-One Update. If we set (31) into (30), we obtain

$$\eta = 1 + [1 + 1/(\hat{a}\hat{c} - 1)]^{1/2}. \tag{33}$$

This choice, proposed by Lukšan in Ref. 3 and studied by Al-Baali in Ref. 23, is different from the choice (30) if $\hat{\gamma}$ is not given by (31). The optimally conditioned scaling parameter is given by (31). The recommended heuristic scaling parameter is $\hat{\gamma} = \hat{c}$.

(U7) Variationally Derived Preconvex Update. See Ref. 13. This update is defined by the choice $\eta = 2$. In this case,

$$H^+ = \gamma\{H + [(\hat{\rho} - 2\hat{a})/b]dd^T + (1/a)[2\hat{a}d - Hy][2\hat{a}d - Hy]^T\}.$$

The optimally conditioned scaling parameter is $\hat{\gamma} = \hat{c}/(2\hat{a}\hat{c} - 1)$. The recommended heuristic scaling parameter is $\hat{\gamma} = \hat{c}$.

(U8) New Heuristic Update. Recently Byrd, Liu, and Nocedal proposed in Ref. 2 a theoretical value

$$\beta = \beta^* + (1/c)(1/v^T G^{-1} v),$$

where

$$v = (1/b)y - (1/c)Bd$$

and G is the exact Hessian matrix. Setting B^+ instead of G, as is usual, we do not obtain any reasonable update. Therefore, we try to approximate G by $(1/\lambda)B$, where λ is some here unknown value. Using the expression for v, we can write

$$v^T G^{-1} v \approx \lambda v^T H v = (\lambda/c)(\hat{a}\hat{c} - 1),$$

which together with the expression for β^* gives

$$\beta = (1/\lambda - 1)/(\hat{a}\hat{c} - 1).$$

From (12), we then obtain

$$\eta = (\lambda\hat{a}\hat{c} - 1)/(\hat{a}\hat{c} - 1).$$

All that remains is to determine the value λ. Since it plays a similar role to the scaling parameter γ in (5), a heuristic value like (22) can be used. We tested the choice $\lambda = (\hat{c}/\hat{a})^r$ for several exponents r and obtained the best results with the value $r = 1/5$. Since the condition $\lambda > 1/\hat{a}\hat{c}$ has to be satisfied, we use the safeguarded value

$$\lambda = \max[\epsilon + 1/\hat{a}\hat{c}, (\hat{c}/\hat{a})^{1/5}]. \tag{34}$$

The final formula is

$$\eta = [\lambda - 1/(\hat{a}\hat{c})]/\max[\epsilon, 1 - 1/(\hat{a}\hat{c})], \tag{35}$$

where $\epsilon > 0$ could be near to the machine precision; we used $\epsilon = 10^{-15}$. The recommended heuristic scaling parameter is $\hat{\gamma} = (\hat{c}/\hat{a})^{1/2}$.

(U9) Preconvex Update Derived from Matrix Factorization. Suppose that the matrix B is factorized in the form

$$B = JJ^T,$$

where J is a regular square matrix. Let

$$B^+ = (J + \Delta J)(J + \Delta J)^T$$

be a similar factorization of the matrix B^+. Consider the problem of finding a regular square matrix ΔJ which minimizes a Frobenius norm $\|J^{-1}\Delta J\|_F$ over all matrices that satisfy the quasi-Newton condition

$$(J + \Delta J)(J + \Delta J)^T d = y.$$

This problem was solved by Zhang and Tewarson in Ref. 25. They prove that such a matrix exists, is unique, and that the matrix

$$B^+ = (J + \Delta J)(J + \Delta J)^T$$

belongs to the preconvex part of the Broyden family. The final solution is (11), where

$$\beta = -1/[(\lambda + 1)^2\hat{c}]$$

and λ is the unique positive root of the equation

$$(\lambda + 1)^2(\hat{c}\lambda^2 - 1) = \lambda^2(\hat{a}\hat{c} - 1). \tag{36}$$

If we use (12), we can determine the parameter η. It is given by the formula

$$\eta = [\hat{c}\lambda^2 - 1/(\hat{a}\hat{c})]/\max[\epsilon, 1 - 1/(\hat{a}\hat{c})], \tag{37}$$

where $\epsilon > 0$ has the same meaning as in (35). The optimally conditioned scaling parameter is $\hat{\gamma} = 1/(\lambda^2\hat{a}\hat{c})$. The recommended heuristic scaling parameter is $\hat{\gamma} = \hat{c}$.

(U10) Update Minimizing the Angle between the Direction Vector and the Negative Gradient. This update was proposed by Zhang and Tewarson in Ref. 26. We do not give the straightforward but formally complicated algebraic derivation here; we only describe the final result. Denote

$$\sigma = 1 - (1/\alpha)(1/\hat{a}), \qquad \tau = 1 - (\hat{c}/\alpha),$$

where α is a steplength [see (1)] and

$$S = \sigma \|d\|^2 - (1/\hat{a})\tau d^T Hy,$$

$$T = \sigma d^T Hy - (1/\hat{a})\tau \|Hy\|^2,$$

$$A = S - (1/\hat{a})T.$$

If $\tau A \neq 0$, then the angle considered has an extremum for

$$1 + \eta(\hat{a}\hat{c} - 1) = \hat{\rho}\alpha\tau S/A,$$

which is either a minimum if $\tau A > 0$ or a maximum if $\tau A < 0$. Since a minimum may not exist, we have to use a safeguard which consists in the bracketing of $1 + \eta(\hat{a}\hat{c} - 1)$ into the interval $[\hat{a}\hat{c}, 2\hat{a}\hat{c}]$; Zhang and Tewarson used the interval $[\hat{a}\hat{c}, 5\hat{a}\hat{c}]$, which gives worse results. The final result is

$$\eta = (2\hat{a}\hat{c} - 1)/(\hat{a}\hat{c} - 1), \tag{38a}$$

if $\tau A \leq 0$, and

$$\eta = \{\max[\hat{a}\hat{c}, \min(2\hat{a}\hat{c}, \hat{\rho}\alpha\tau S/A)] - 1\}/(\hat{a}\hat{c} - 1), \tag{38b}$$

otherwise. The recommended heuristic scaling parameter is $\hat{\gamma} = (\hat{c}/\hat{a})^{1/2}$.

(U11) Update Minimizing the Norm of the Direction Vector. As in the previous case, the norm of the next direction vector depends on the parameter η. We can try to find a value of the parameter η which minimizes the norm

$$\|s^+\| = \|H^+ g^+\|.$$

This minimum always exists if $A \neq 0$, and it is reached for

$$1 + \eta(\hat{a}\hat{c} - 1) = \hat{\rho}\alpha\tau S/A,$$

where now $\tau = 1 - (\hat{c}/\alpha)$ and

$$S = \|d\|^2 - (1/\hat{a}) d^T Hy,$$

$$T = d^T Hy - (1/\hat{a})\|Hy\|^2,$$

$$A = S - (1/\hat{a})T.$$

Using a safeguard as in the previous case, we obtain a new formula,

$$\eta = \{\max[\hat{a}\hat{c}, \min(2\hat{a}\hat{c}, \alpha\tau\hat{\rho}S/A)] - 1\}/(\hat{a}\hat{c} - 1). \tag{39}$$

The recommended heuristic scaling parameter is $\hat{\gamma} = (\hat{c}/\hat{a})^{1/2}$.

(U12) Least Prior Deviation Update. This update, proposed by Mifflin and Nazareth in Ref. 27, is based on the minimization of the last prior deviation $(H^+y^- - d^-)G(H^+y^- - d^-)$, where $d^- = x - x^-$, $y^- = g - g^-$ are the most recent differences and G is a symmetric positive-definite matrix such that $Gd = y$. After some algebraic manipulations, we can derive that the last prior deviation is minimal if

$$\eta = 1/[1 - \hat{a}(d^Ty^-/y^Td^-)]. \tag{40}$$

This choice has properties very similar to the rank-one update [for the quadratic function, we have $d^Ty^- = y^Td^-$, so that $\eta = 1/(1 - \hat{a})$ as in (30)] and it is necessary to use some safeguard. We again use (40) only if $\hat{a}(d^Ty^-/y^Td^-) < 1$ and $\eta = 1$ (BFGS method) otherwise. The recommended heuristic scaling parameter is $\hat{\gamma} = (\hat{c}/\hat{a})^{1/2}$.

4. Implementation of Variable Metric Methods

A very important part of the iteration (1)–(5) is the line search procedure. We describe in detail our version of this algorithm:

Line Search Algorithm. Given $\bar{\Delta} > 0$, $0 < \epsilon_1 < 1/2$, and $\epsilon_1 < \epsilon_2 < 1$. We recommend $\epsilon_1 = 0.0001$ and $\epsilon_2 = 0.9$. The parameter $\bar{\Delta}$ is determined from the problem description and the value $\bar{\Delta} = 1000$ can be the default.

Step 1. Determine the initial estimate $\alpha_1 = \min(1, 4(\underline{F} - F)/s^Tg)$ of α^+, where $\underline{F}$ is the lower bound for the function value. Set $\bar{\alpha}_1 := 0$. Set $i := 1$.

Step 2. Set $\alpha_i := \min(\alpha_i, \bar{\Delta}/\|s\|)$. Set $\underline{\alpha}_i := \bar{\alpha}_i$ and $\bar{\alpha}_i := \alpha_i$. If the conditions (3) and (4) are satisfied, with F^+ and g^+ replaced by $F(x + \alpha_i s)$ and $g(x + \alpha_i s)$ respectively, then set $\alpha^+ := \alpha_i$ and terminate the computation. If both (3) and $s^Tg(x + \alpha_i s) < 0$ hold, then go to Step 3; otherwise, go to Step 4.

Step 3. If either $\alpha_i = \bar{\Delta}/\|s\|$ or $i > 10$, then set $\alpha^+ = \bar{\alpha}_i$ and terminate the computation; otherwise, determine the new estimate $\alpha_i > \bar{\alpha}_i$ by a safeguarded cubic extrapolation. Set $\alpha_i := \max(\alpha_i, 1.1\bar{\alpha}_i)$. Set $\alpha_i := \min(\alpha_i, 10^3\bar{\alpha}_i)$. Set $i := i + 1$, and go to Step 2.

Step 4. If $i > 10$, then set $\alpha^+ = \underline{\alpha}_i$ and terminate the computation; otherwise, determine the new estimate $\underline{\alpha}_i < \alpha_i < \bar{\alpha}_i$ by a safeguarded cubic interpolation. If $\underline{\alpha}_i = 0$, then set $\alpha_i := \max(\alpha_i, \underline{\alpha}_i + 10^{-2}(\bar{\alpha}_i - \underline{\alpha}_i))$; otherwise, set $\alpha_i := \max(\alpha_i, \underline{\alpha}_i + 10^{-1}(\bar{\alpha}_i - \underline{\alpha}_i))$. Set $\alpha_i := \min(\alpha_i, \underline{\alpha}_i + 0.9(\bar{\alpha}_i - \underline{\alpha}_i))$.

Step 5. If the conditions (3) and (4) are satisfied, with F^+ and g^+ replaced by $F(x + \alpha_i s)$ and $g(x + \alpha_i s)$ respectively, then set $\alpha^+ := \alpha_i$ and terminate the computation. If both (3) and $s^T g(x + \alpha_i s) < 0$ hold, then set $\underline{\alpha}_i := \alpha_i$; otherwise, set $\bar{\alpha}_i := \alpha_i$. Set $i := i + 1$ and go to Step 4.

Another important part of our optimization algorithm is the termination criterion. We terminate computation whenever

$$\|g^+\| \le 10^{-6},$$

or whenever

$$\|x^+ - x\|/\max(1, \|x^+\|) \le 10^{-16},$$

or whenever

$$|F^+ - F|/\max(1, |F^+|) \le 10^{-16}$$

holds in two consecutive iterations, or whenever

$$F \le \underline{F},$$

where $\underline{F}$ is the lower bound for the function value. These termination criteria have different significance. We consider the first criterion, $\|g^+\| \le 10^{-6}$, as the basic one. The other critiera were active only rarely in our computations. More details about this fact are given in the next section.

5. Computational Experiments

This section contains a great number of experiments with different choices of the parameters $\rho > 0$, $\gamma > 0$, and η in (5). Updates with 12 different values of the parameter η are distinguished by the notation U1, ..., U12 according to the numbering given in Section 3; U1 is the BFGS method. All tests were made using 25 test problems taken from Refs. 3 and 4, specifically:

(F1) Chained Rosenbrock function.
(F2) Chained Wood function.

(F3) Chained Powell singular function.
(F4) Chained Cragg and Levy function.
(F5) Generalization of the Broyden tridiagonal function.
(F6) Generalization of the Broyden banded function.
(F7) Toint 7-diagonal function.
(F8) Nazareth trigonometric function.
(F9) Toint trigonometric function.
(F10) Penalty function.
(F11) Augmented Lagrangian function.
(F12) Generalization of the first Brown function.
(F13) Generalization of the second Brown function.
(F14) Discrete boundary-value problem.
(F15) Discretization of a variational problem.
(F16) Extended Rosenbrock function.
(F17) Extended Powell singular function.
(F18) First penalty function.
(F19) Second penalty function.
(F20) Variably dimensioned function.
(F21) Trigonometric function.
(F22) Brown almost-linear function.
(F23) Discrete boundary-value function.
(F24) Discrete integral equation function.
(F25) Modification of the Broyden tridiagonal function.

Problems F16–F25 are exactly Problems 21–30 from Ref. 4.

We hope that this set is sufficiently large and that the individual tests are sufficiently difficult, diversified, and suitable for testing variable metric methods. Therefore, we always give only summary results over all 25 test problems. The tables in this section contain the total numbers of iterations and function evaluations. Most tests were made with 20 variables ($N = 20$).

The first tests demonstrate the influence of the value $\rho > 0$ on the efficiency of selected variable metric updates for both preliminary scaling (PS) and controlled scaling (CS) strategies. Similar results were obtained also for the remaining four values of the parameter η. The results in Table 1 show that the choices (16) or (18) and (19) are usually better than the simple choice $\rho = 1$. The value (16) is advantageous together with preliminary scaling, while the values (18) and (19) are superior when controlled scaling is used; the difference between (18) and (19) is caused by roundoff errors. We recommend the choice (19) and use the notation $\rho \neq 1$ for this case in the subsequent text.

Now, we demonstrate the influence of the value $\gamma > 0$ on the efficiency of selected variable metric updates for both controlled scaling (CS) and

Table 1. Influence of stabilization, $N = 20$.

Update	Scaling	$\rho = 1$	(16)	(17)	(18)	(19)
U1	PS	1904–2120	1631–1810	1917–2113	1712–1914	1841–2057
U1	CS	1180–1445	1114–1364	1276–1585	1054–1275	1056–1272
U3	PS	2706–2876	2416–2596	2406–2551	2423–2586	2383–2541
U3	CS	1207–1378	1259–1427	1257–1396	1211–1353	1206–1353
U4	PS	1423–1734	1359–1671	1375–1665	1323–1636	1367–1674
U4	CS	1192–1435	1120–1357	1203–1434	1076–1296	1086–1316
U6	PS	1498–1791	1366–1679	1501–1781	1430–1728	1437–1758
U6	CS	1377–1661	1319–1604	1387–1671	1302–1576	1310–1572
U7	PS	1616–1891	1397–1634	1520–1729	1456–1729	1454–1725
U7	CS	1097–1364	1190–1482	1319–1640	1113–1380	1135–1391
U8	PS	1625–1797	1516–1715	1718–1931	1590–1803	1558–1760
U8	CS	1159–1353	1082–1274	1155–1381	1099–1285	1071–1262
U9	PS	1678–1917	1503–1752	1715–1935	1553–1805	1591–1824
U9	CS	1075–1350	1087–1349	1212–1495	1022–1290	1019–1278
U12	PS	1397–1646	1421–1685	1499–1743	1342–1598	1408–1698
U12	CS	1178–1425	1160–1419	1390–1648	1085–1323	1081–1314

Table 2. Influence of scaling, $N = 20$.

Update	Scaling	Recommended	(24)	(26)
U1	CS	1180–1445	1253–1436	1580–1800
U1	AS	1320–1618	2297–2478	F (12)
U3	CS	1207–1378	1207–1378	1333–1547
U3	AS	1362–1511	1362–1511	1398–1747
U4	CS	1192–1435	1307–1575	F (1)
U4	AS	1717–1950	F (5)	F (5)
U6	CS	1340–1730	1407–1712	1361–1623
U6	AS	1684–2088	F (4)	F (2)
U7	CS	1097–1363	1373–1612	1373–1612
U7	AS	1511–1836	F (4)	F (4)
U8	CS	1159–1353	1487–1679	1378–1595
U8	AS	1689–1975	F (1)	F (4)
U9	CS	1075–1350	1307–1550	1391–1630
U9	AS	1428–1765	F (1)	F (5)
U12	CS	1178–1425	1222–1461	1420–1691
U12	AS	1968–2299	F (3)	F (2)

scaling in all iterations (AS). We always assume the value $\rho = 1$. Recommended values $\gamma > 0$ for individual variable metric updates are proposed in Section 3. Four values of the parameter η are excluded, since the DFP update U2 is inefficient and it is not possible to determine simple variationally derived scaling parameters for the updates U5, U10, U11.

Table 3a. Test results, $N = 20$.

Update	Parameter	NS	PS	CS	AS
U1	$\rho = 1$	2543–3341	1904–2120	1180–1445	1320–1618
U1	$\rho \neq 1$	2457–3249	1841–2057	1056–1272	1291–1577
U2	$\rho = 1$	F (8)	F (12)	3297–3537	1580–1841
U2	$\rho \neq 1$	F (9)	F (13)	2444–2464	1672–1941
U3	$\rho = 1$	3030–3646	2706–2876	1207–1378	1362–1511
U3	$\rho \neq 1$	2520–3068	2384–2541	1206–1353	1384–1534
U4	$\rho = 1$	2438–3359	1423–1734	1192–1435	1717–1950
U4	$\rho \neq 1$	2183–3041	1367–1674	1086–1316	1570–1786
U5	$\rho = 1$	2486–3015	1791–1992	1275–1455	1349–1503
U5	$\rho \neq 1$	2630–3217	1614–1780	1224–1361	1358–1513
U6	$\rho = 1$	2018–2911	1462–1796	1340–1730	1684–2088
U6	$\rho \neq 1$	2110–3016	1399–1723	1159–1508	1619–1963
U7	$\rho = 1$	2324–3251	1616–1891	1097–1364	1511–1836
U7	$\rho \neq 1$	2222–3087	1454–1725	1135–1391	1458–1789
U8	$\rho = 1$	2396–3125	1625–1797	1159–1353	1689–1975
U8	$\rho \neq 1$	2357–3075	1558–1760	1071–1262	1618–1919
U9	$\rho = 1$	2217–3043	1678–1917	1075–1350	1428–1765
U9	$\rho \neq 1$	2288–3123	1591–1824	1019–1278	1314–1639
U10	$\rho = 1$	2254–3096	1401–1676	1213–1478	1772–2057
U10	$\rho \neq 1$	2175–3012	1334–1596	1069–1332	1900–2194
U11	$\rho = 1$	2868–3726	1885–2094	1156–1361	1811–2035
U11	$\rho \neq 1$	2626–3431	1650–1855	1106–1310	1631–1887
U12	$\rho = 1$	2361–3223	1397–1646	1178–1425	1968–2299
U12	$\rho \neq 1$	2069–2778	1408–1698	1081–1314	1889–2247

The notation F (number of fails) means that a given number of runs failed (more than 1000 iterations or 2000 function evaluations). Table 2 shows that variable metric updates with recommended heuristic scaling parameters are usually much better than those with optimally conditioned or variationally derived scaling parameters especially for scaling in all iterations.

Now, we are in the position to present basic tests which demonstrate the efficiency of 12 individual variable metric updates with four scaling strategies making use of recommended heuristic scaling parameters: no scaling (NS), preliminary scaling (PS), controlled scaling (CS), scaling in all iterations (AS), and with two choices, $\rho = 1$ and $\rho \neq 1$; see (19). Again, we use the 25 test problems listed above with 20 variables (Table 3a), 40 variables (Table 3b), and 80 variables (Table 3c); we present only total results. The last two tables do not contain the extremal strategies NS and AS. The notation F (number of fails) has the same meaning as above.

Table 3b. Test results, $N = 40$.

Update	Parameter	PS	CS
U1	$\rho = 1$	2669–2917	1540–1848
U1	$\rho \neq 1$	2606–2855	1502–1870
U2	$\rho = 1$	F (13)	2712–3094
U2	$\rho \neq 1$	F (10)	2686–3070
U3	$\rho = 1$	3568–3699	1512–1663
U3	$\rho \neq 1$	3450–3587	1510–1645
U4	$\rho = 1$	1949–2292	1488–1765
U4	$\rho \neq 1$	1845–2183	1471–1740
U5	$\rho = 1$	2627–2830	1612–1785
U5	$\rho \neq 1$	2365–2539	1632–1786
U6	$\rho = 1$	2106–2562	1651–2089
U6	$\rho \neq 1$	2001–2443	1521–1896
U7	$\rho = 1$	2196–2551	1635–2024
U7	$\rho \neq 1$	2171–2513	1476–1829
U8	$\rho = 1$	2325–2508	1484–1690
U8	$\rho \neq 1$	2270–2491	1461–1634
U9	$\rho = 1$	2457–2765	1545–1891
U9	$\rho \neq 1$	2380–2678	1481–1851
U10	$\rho = 1$	2073–2428	1705–2062
U10	$\rho \neq 1$	1995–2344	1592–1960
U11	$\rho = 1$	2607–2813	1502–1756
U11	$\rho \neq 1$	2399–2634	1584–1839
U12	$\rho = 1$	1935–2273	1618–1890
U12	$\rho \neq 1$	1903–2242	1510–1813

Before formulating our conclusions, we have to give several comments on the correctness of the proposed results. First, we have to make clear the role of different termination criteria mentioned in the previous section. In fact, the gradient norm criterion was used in almost all the cases. There were only two exceptions. The first exception appeared when Problem 9 was solved. This problem has a relatively large gradient norm in the neighborhood of the solution. We obtained the solution with 8 significant digits in all variables, but the gradient norm lay between 10^{-6} and 10^{-4} (it depended on the number of variables). The second exception was connected with Problem 15. This function has a bounded local minimum, but it is unbounded in R^n. For $n = 40$ and $n = 80$, we always obtained unbounded value less than $\underline{F} = -10^{60}$. For small n, we found sometimes an unbounded value when the NS strategy was used. Since the number of iterations for reaching an unbounded value was slightly less than that for reaching a local minimum, this phenomenon could not distort our comparison (it slightly improved the results for the strategy which was shown as

Table 3c. Test results, $N = 80$.

Update	Parameter	PS	CS
U1	$\rho = 1$	3645–4063	2190–2634
U1	$\rho \neq 1$	3323–3667	2026–2494
U2	$\rho = 1$	F (13)	F (4)
U2	$\rho \neq 1$	F (11)	F (1)
U3	$\rho = 1$	5263–5449	2083–2286
U3	$\rho \neq 1$	4532–4727	2094–2304
U4	$\rho = 1$	2733–3259	2173–2464
U4	$\rho \neq 1$	2511–2991	2041–2340
U5	$\rho = 1$	3502–3756	2710–2956
U5	$\rho \neq 1$	3329–3559	2804–3041
U6	$\rho = 1$	2963–3617	2573–3164
U6	$\rho \neq 1$	2859–3555	2439–3066
U7	$\rho = 1$	3182–3702	2322–2874
U7	$\rho \neq 1$	2944–3490	2392–3039
U8	$\rho = 1$	3159–3421	2007–2266
U8	$\rho \neq 1$	3096–3624	1901–2233
U9	$\rho = 1$	3471–3982	2395–2906
U9	$\rho \neq 1$	3260–3748	2173–2649
U10	$\rho = 1$	2807–3279	2329–2760
U10	$\rho \neq 1$	2722–3189	2120–2660
U11	$\rho = 1$	3464–3780	2157–2398
U11	$\rho \neq 1$	3156–3459	2071–2358
U12	$\rho = 1$	2652–3127	2230–2634
U12	$\rho \neq 1$	2594–3058	2062–2449

the worst one). Besides different termination criteria, we have to consider also the situation with different solution points. For example, we obtained different local minima (with gradient norm less than 10^{-6}) when Problem 2 was solved. But detailed comparison of these cases shows that there was no statistically significant dependence between the reached local minimum and the number of iterations needed for finding it.

According to the results presented in Tables 3a–Table 3c and the comments stated above, we can express several conclusions, which of course hold only for our collection of test problems:

(C1) Controlled scaling improves considerably the efficiency of variable metric updates especially for large sparse problems.

(C2) Usually, better results are obtained with the choice $\rho \neq 1$ given by (19). The proportional improvement is greater for small dense problems.

(C3) Updates belonging to the convex part of the Broyden family are very sensitive to scaling, and their improvement by controlled scaling is remarkable. This effect is very apparent in the case where the DFP method is used.

(C4) Updates belonging to the preconvex part of the Broyden family are more efficient than other variable metric updates when scaling is not used. However, their improvement by controlled scaling is slightly lower.

(C5) Suitable choices for small dense problems are either U4 (the safeguarded rank-one update) or U12 (the least prior deviation update). These choices deteriorate less if scaling is not used.

(C6) A very interesting conclusion is the high efficiency of the choice U3 (Hoshino update) with controlled scaling for large sparse problems.

(C7) Our heuristic update U8 is very efficient especially if the controlled scaling is used. This update is more stable than the Hoshino update U3 when preliminary scaling is used. It is a very suitable update for large sparse problems. This property is probably conditioned by the fact that this update can belong to both the convex and the preconvex part of the Broyden family.

The above conclusions are conditioned by the test problems used. The advantage of our test problems is that they allow one to increase the number of variables. We also made experiments with 30 standard test problems given in Ref. 4; the last 11 of them were considered with six variables. These problems, which are small, dense, and usually formulated as problems of nonlinear regression, indicated slightly different results. The main difference consisted in the fact that the efficiency of variable metric methods was not improved by scaling. This fact probably gave rise to the widely held notion that scaling of variable metric updates is not advantageous. But this is not always the case, as we have shown in this paper. However, other conclusions given above remain unchanged also for test problems given in Ref. 4. Again, the choice $\rho \neq 1$ given by (19) increased the efficiency of variable metric updates, and the choices U4 (safeguarded rank-one update) and U12 (least prior deviation update) considerably outperformed the BFGS update.

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